The Geography of Disparity in Diabetes Clinical Trial Access

JSC370 Final Project — Deliverable Report

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Interactive figures. On the Interactive Visualizations page, Figures 1, 2, and 3 are interactive versions of this report's state choropleth (Figure 1) and county distance histogram (Figure 3), plus an interactive scatter of county prevalence versus trial distance. Full source and pipeline: https://github.com/shifosss/JSC370-Midterm-Proj.

1 Introduction

1.1 Background and Motivation

Diabetes mellitus represents a major public health crisis in the United States, exhibiting profound geographic and socioeconomic disparities (Barker et al., 2011). Socioeconomic status — including poverty, education, and health insurance — plays a critical role in disease prevalence and management (Hill-Briggs et al., 2021). As the medical industry continues to develop new Type 2 diabetes therapies, a paradox remains: the opportunities to participate in clinical trials are not distributed equitably across the national geography. Trial locations are frequently driven by institutional infrastructure and sponsor logistics, creating a spatial mismatch that systematically excludes vulnerable, rural, and high-burden populations from research (Tanner et al., 2015; Unger et al., 2019). This gap manifests as both physical travel barriers and a lack of local trial availability, directly limiting the generalizability of clinical research (Kirkwood et al., 2024).

1.2 Research Questions and Hypotheses

This project addresses two related but distinct aims:

1. **Descriptive aim.** To what extent is the geographic distribution of U.S. Type 2 diabetes clinical trial sites aligned with the spatial pattern of disease burden? Where are the largest gaps between trial availability and local need?
2. **Modeling aim.** Does including clinical trial access measures (local trial density and distance to the nearest trial site) improve prediction of county-level diabetes prevalence beyond a model that uses only socioeconomic and demographic characteristics, and which features carry the most predictive signal?

We hypothesize that (1) trial sites are spatially concentrated in socioeconomically advantaged counties with established research infrastructure rather than in the highest-burden areas, and (2) adding trial-access variables to an SES-only prediction model will yield little improvement in cross-validated accuracy, consistent with trial access being a redundant predictor once socioeconomic structure is already in the feature set.

2 Methods

2.1 Data Acquisition

The full dataset was assembled exclusively via programmatic API pulls to compile a multi-dimensional, county-level dataset of the United States:

1. **Clinical trial access** — ClinicalTrials.gov v2 API: all U.S. Type 2 diabetes studies ever registered (no date filter), a cumulative snapshot of where trial infrastructure has been placed.
2. **Disease burden** — CDC PLACES: county-level age-adjusted diabetes prevalence and related population-health measures (2022 release cycle; underlying BRFSS data 2020–2022).
3. **Socioeconomic context** — American Community Survey 5-Year 2022 estimates (pooling 2018–2022) for county and state covariates including poverty, income, education, insurance, and demographic composition.
4. **Healthcare infrastructure** — NPI Registry: endocrinologist counts and academic medical center presence by county.

Supplementary contextual variables came from the 2020 Census Decennial (rural population share by state) and a hardcoded lookup of Medicaid expansion status as of January 2024. The 2022 Census Gazetteer and FCC Census API were used for geocoding. All sources are cross-sectional; temporal misalignment across vintages is modest because structural county characteristics change slowly relative to the one- to two-year offsets between sources.

2.2 Data Cleaning and Wrangling

The raw nested JSON payloads from ClinicalTrials.gov were flattened into separate trial-level and site-level records. Because the clinical-trial data encoded locations as city/state strings rather than FIPS identifiers, the 2022 Census Gazetteer combined with the FCC Census API (reverse geocoding latitude/longitude to FIPS) was used to map each trial site to a 5-digit county FIPS code. To measure geographic isolation, straight-line distances were computed using the Haversine formula and `scipy.spatial.cKDTree` from each county's population centroid to the nearest trial site. All datasets were merged on the county FIPS code. Missing values for core modeling features were handled via complete-case analysis, and highly skewed variables (total population, trial distance, and trial density) were log-transformed (`log1p` or `log`) to stabilize variance.

2.3 Analytical Tools and Modeling

EDA and feature engineering used pandas and numpy; static visualizations were produced with seaborn and matplotlib, and the interactive figures on the companion Visualizations page with Plotly. The response variable for Aim 2 is **county-level age-adjusted diabetes prevalence** (`places_diabetes` from CDC PLACES). The 13 predictors fall into two groups: *trial-access variables* (log trial density, log distance to the nearest site) and *socioeconomic/demographic covariates* (poverty rate, median household income, Gini index, uninsured rate, unemployment rate, bachelor's-degree attainment, vehicle access, internet access, % non-Hispanic Black, % Hispanic, and log population).

Baseline vs. augmented prediction comparison. To isolate the predictive contribution of trial-access features, each model was fit in two configurations: an *SES-only baseline* using the 11 socioeconomic/demographic predictors, and a *Full* model adding the two trial-access predictors. Comparing cross-validated R^2, RMSE, and MAE between configurations quantifies the incremental lift from trial-access features. Three model families were used:

- **Elastic Net** (scikit-learn `ElasticNetCV`; 5-fold CV over five `l1_ratio` values and 100 `alpha` values; predictors standardized) — a regularized linear model whose coefficient magnitudes indicate feature importance under linearity.
- **Random Forest** (`RandomForestRegressor` tuned via 30-iteration randomized search over trees, depth, leaves, and feature fractions) — a non-linear ensemble; permutation importance ranks features by contribution to out-of-sample accuracy.
- **XGBoost** (`XGBRegressor` tuned via 30-iteration randomized search over n_estimators, learning rate, depth, subsample, and column subsample) — gradient-boosted trees that typically achieve the highest predictive accuracy.

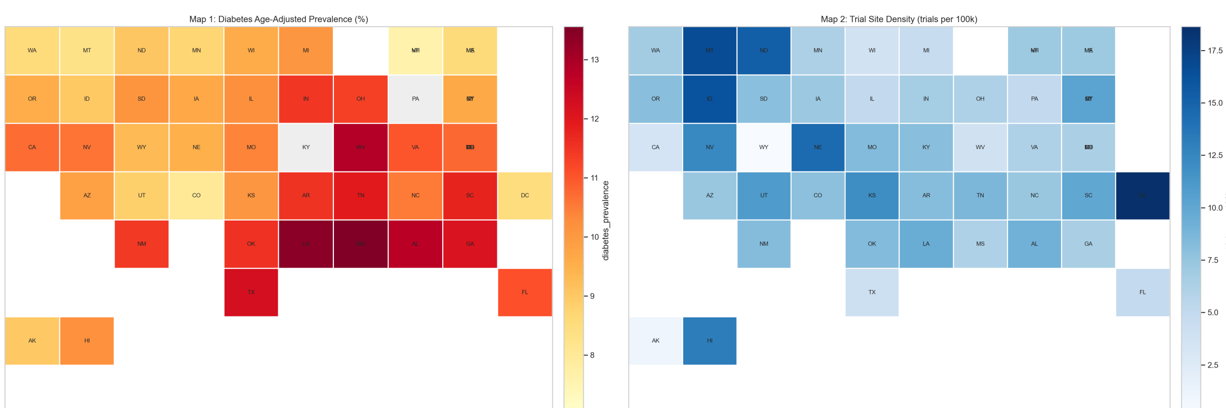
All three models were evaluated under the same 5-fold cross-validation scheme (shuffled, `random_state=42`). **Feature importance** was assessed with SHAP values from the best-performing model (XGBoost), permutation importance from the tuned Random Forest, and standardized coefficients from Elastic Net.

3 Results

3.1 Descriptive: State and County Access Gaps (Aim 1)

The data pipeline aggregated 3,646 U.S. Type 2 diabetes studies and 47,118 site records. Because 73.4% of U.S. counties host no diabetes trial site at all, state-level summaries are the primary descriptive lens; county-level results supplement them by quantifying the within-state variation that state aggregates mask.

3.1.1 State-Level Burden and Trial Density Do Not Line Up Cleanly



(a) Diabetes age-adjusted prevalence (%).

(b) Trial-site density (trials per 100k population).

Figure 1: **Figure 1.** State tile maps: diabetes prevalence (left) vs. trial density (right). The diabetes belt in the South and Appalachia is visible in the left panel, while the trial-density map is more fragmented and includes several smaller, infrastructure-rich states that do not rank among the highest-burden areas.

To formalize the mismatch, a state-level *coverage residual* was computed: observed trial density minus the trial density expected for a state's diabetes-burden decile. Negative values indicate under-coverage relative to peer states with similar burden; positive values indicate over-coverage.

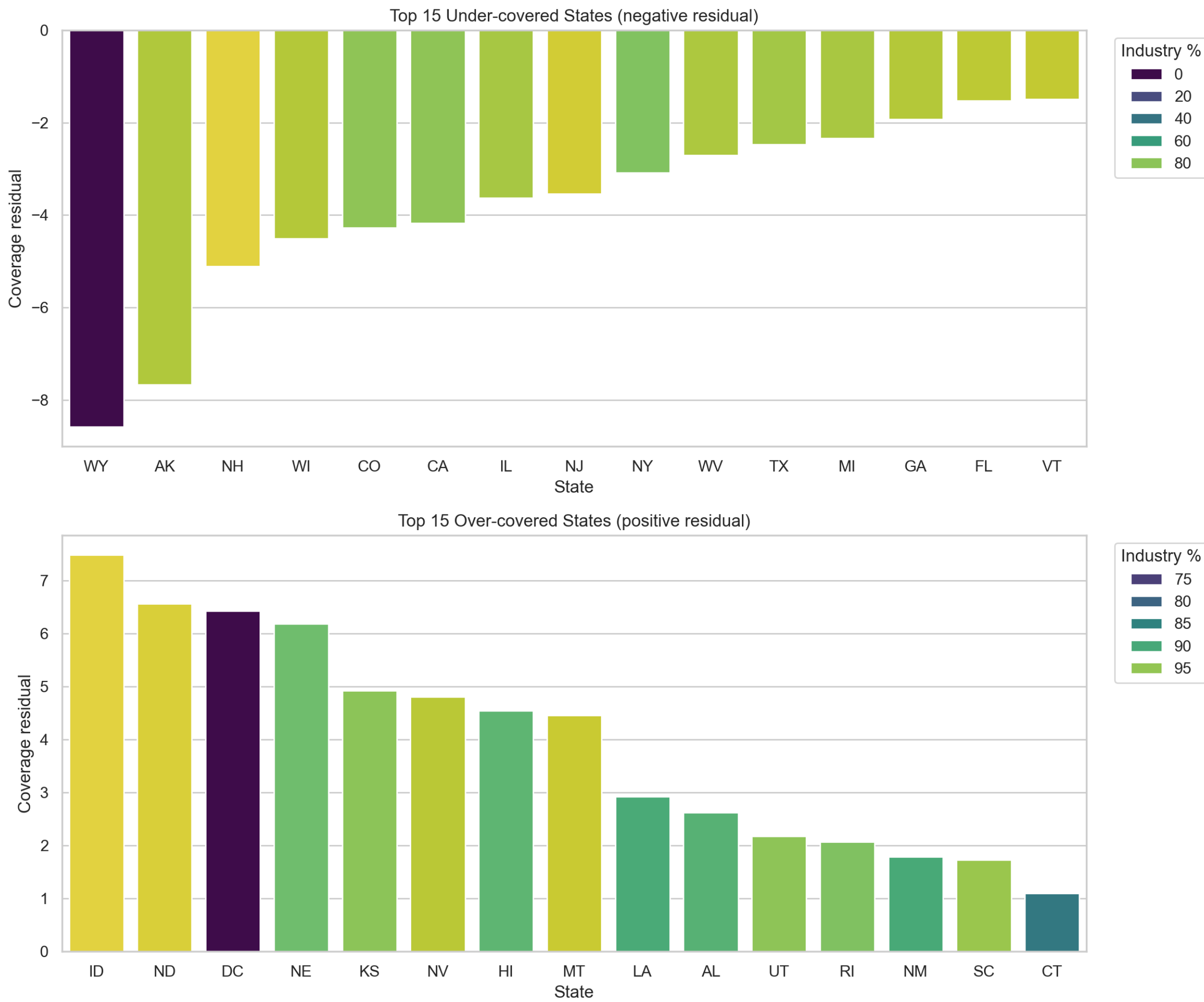


Figure 2: Coverage residual outliers by state. Each bar is a state's coverage residual (observed trial density minus the mean density for its diabetes-burden decile). Bars are colored by the percentage of the state's trials that are industry-sponsored.

Figure 2. Coverage residual by state. The residual analysis sharpens the map comparison. Several high-burden states are under-covered relative to peers in the same burden decile, while some Mountain Plains and small-jurisdiction states are over-covered. Sponsor composition is concentrated: nationally, industry-sponsored trial density is 5.39 trials per 100k population versus 0.78 for non-industry trials — industry density is about 6.9× higher. This pattern is consistent with site placement being influenced by sponsor logistics and institutional presence.

Table 1. Five most under-covered states relative to diabetes burden.

State	Trial count	Site count	Trials per 100k	Coverage residual	Industry share (%)
WY	1	1	0.17	-8.58	0.0
AK	8	9	1.09	-7.66	87.5
NH	98	103	7.10	-5.11	96.9
WI	221	285	3.76	-4.50	88.2
CO	458	685	7.94	-4.28	80.8

Table 2. Five most over-covered states relative to diabetes burden.

State	Trial count	Site count	Trials per 100k	Coverage residual	Industry share (%)
ID	301	376	16.23	7.48	99.7
ND	119	132	15.32	6.57	99.2
DC	125	136	18.64	6.43	70.4
NE	283	404	14.45	6.19	92.6
KS	351	475	11.96	4.92	94.6

Together, Figures 1–2 and Tables 1–2 show that trial availability is not uniformly low or high across states; rather, there is substantial variation in how well trial density aligns with burden.

3.1.2 County-Level Access Gaps Are Large

State aggregates mask within-state variation. Of 3,221 counties, only 856 (26.6%) host at least one diabetes trial site; the other 2,365 (73.4%) have none. County-level trial density is therefore zero-inflated, and county modeling (Aim 2) relies on the distance-to-nearest-site measure rather than local trial counts to capture access variation.

Table 3. County-level access summary.

Metric	Value
Counties in final county file	3,221
Counties with at least one trial site	856 (26.6%)
Counties without a trial site	2,365 (73.4%)
Median nearest-site distance among no-site counties	58.4 km
Share of no-site counties more than 50 km away	59.7%
Share of no-site counties more than 100 km away	24.0%
Share of no-site counties more than 200 km away	8.2%

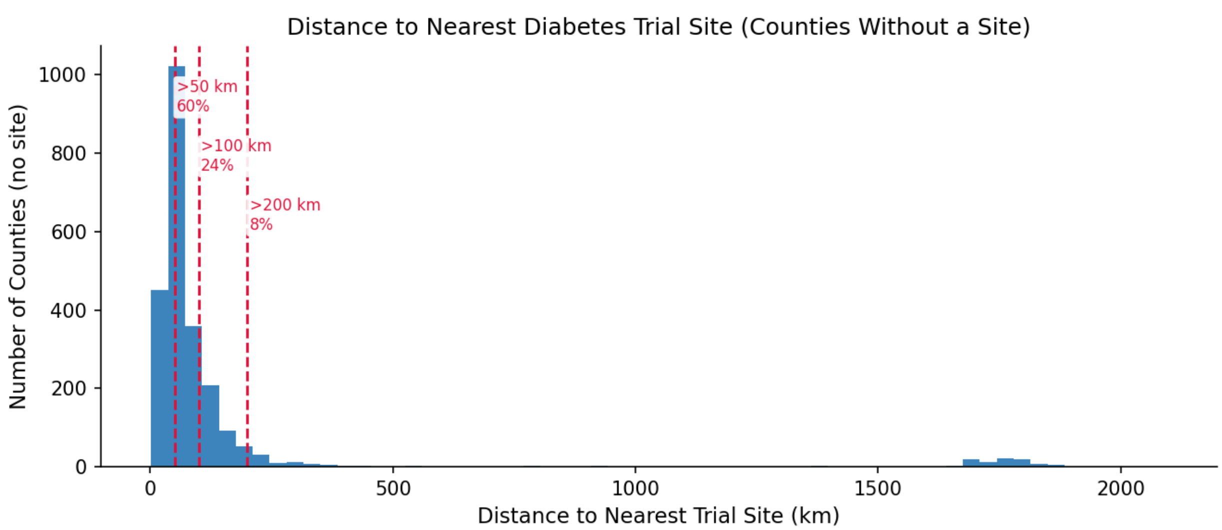


Figure 3: Distance to the nearest diabetes trial site for counties without a local site.

Figure 3. County-level distance distribution. For no-site counties, travel distance is often substantial. A state can look moderately well covered in aggregate while still leaving most of its counties without feasible local trial access. An interactive version with a per-state filter is available on the Visualizations page.

Among counties with at least one trial site, endocrinologist density and trial density co-occur — both likely reflect academic medical centers and large health systems. The bivariate scatter (available on the Visualizations page) is descriptive only; the Aim 2 models test whether these features contribute independent predictive signal.

3.2 Modeling: Trial-Access Features Add Little Predictive Lift (Aim 2)

Each of the three model families (Elastic Net, Random Forest, XGBoost) was fit twice under a 5-fold shuffled CV scheme: once with the 11 SES/demographic predictors only, and once with the same features plus the two trial-access variables (`log_trials`, `log_dist`). The incremental lift from trial-access features is the difference in CV R^2 (and RMSE/MAE) between the two configurations.

Table 4. Cross-validated performance — SES-only baseline vs. Full model (adds `log_trials` and `log_dist`).

Model	R^2 (SES-only)	R^2 (Full)	ΔR^2 (Full – SES-only)	RMSE (SES-only)	RMSE (Full)	MAE (SES-only)	MAE (Full)
Elastic Net	0.8618	0.8638	+0.0020	0.8655	0.8591	0.6617	0.6557
Random Forest	0.8800	0.8821	+0.0021	0.8064	0.7994	0.6113	0.6055
XGBoost	0.8886	0.8916	+0.0030	0.7771	0.7665	0.5877	0.5775

Table 4 reports the CV metrics for the six model–configuration combinations. Across all three model families the Full model's R^2 improvement over the SES-only baseline is modest in absolute terms; the trial-access variables do not substantially reduce cross-validated prediction error. The gradient-boosted model achieves the highest R^2 of the three, and the relative ranking (XGBoost Random Forest > Elastic Net) is stable across configurations, consistent with the non-linear models capturing interactions among SES predictors that linear regularization cannot.

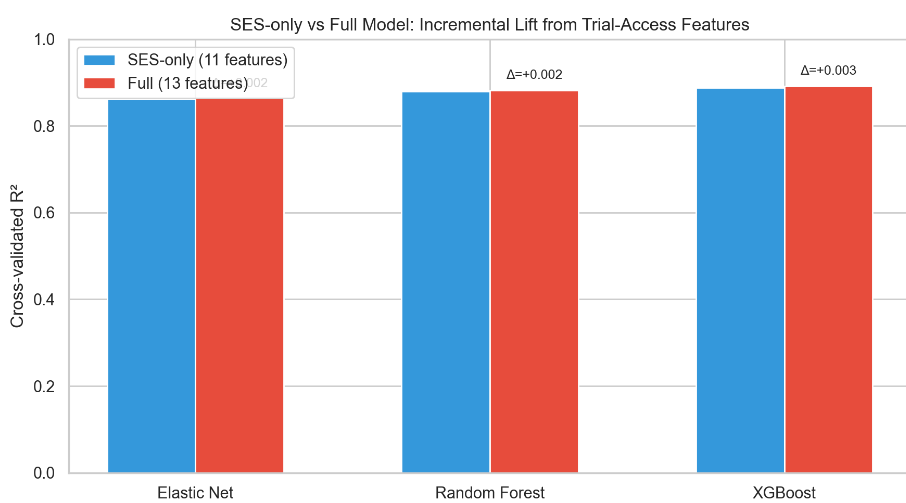
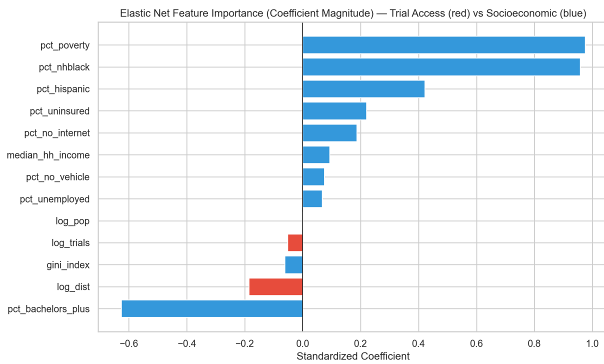


Figure 4: SES-only vs. Full model R^2 for Elastic Net, Random Forest, and XGBoost, with Δ annotations.

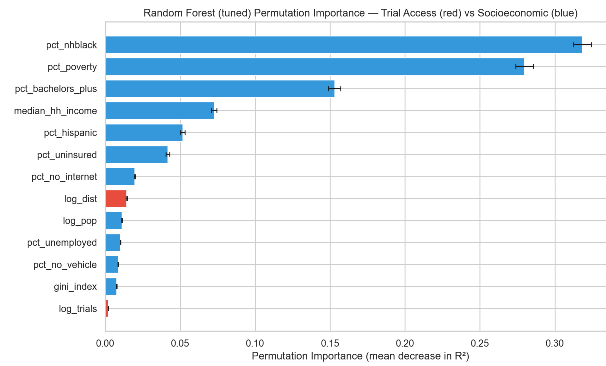
Figure 4. Grouped-bar comparison of cross-validated R^2 for each model family in the SES-only (blue) and Full (red) configurations. The annotated Δ values make the incremental lift explicit.

3.2.1 Feature Importance

The feature-importance analyses tell a consistent story. Regardless of which lens is used — standardized Elastic Net coefficients, Random Forest permutation importance, or SHAP values from XGBoost — socioeconomic and demographic variables dominate, and the two trial-access features sit near the bottom of the ranking.

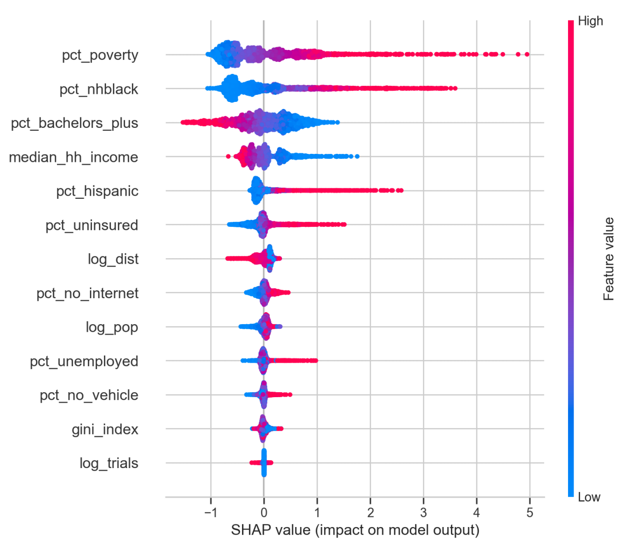


(a) Elastic Net standardized coefficients (red: trial-access; blue: SES/demographic).

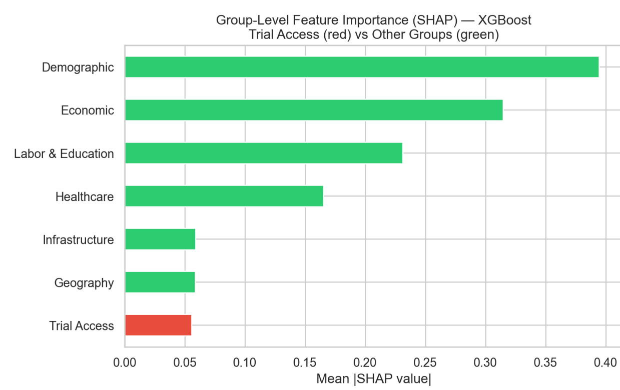


(b) Random Forest permutation importance (mean decrease in CV R^2).

Figure 5: **Figure 5.** Feature importance from the linear (left) and tree-ensemble (right) models. Both rankings place the trial-access variables near the bottom; poverty, % non-Hispanic Black, % Hispanic, median household income, and insurance status carry the largest weights.



(a) SHAP summary for XGBoost (per-county feature contributions).



(b) Mean |SHAP value| aggregated by predictor group — Trial Access (red) vs. SES/demographic groups (green).

Figure 6: **Figure 6.** SHAP decomposition of XGBoost predictions. The summary plot (left) places the structural SES predictors — % non-Hispanic Black, poverty, income, % Hispanic — at the top. The group-aggregated view (right) shows the trial-access block contributing an order of magnitude less than Economic or Demographic groups.

Taken together, Figures 4–6 and Table 4 support Hypothesis 2: **trial-access features add little predictive signal for county-level diabetes burden once socioeconomic and demographic structure is already in the model.** The Full-model ΔR^2 values are small enough to be consistent with CV-split noise, and every feature-importance lens places trial access below every major SES block.

4 Conclusions and Summary

Hypothesis outcomes. The descriptive evidence (Aim 1) supports Hypothesis 1. State-level trial density does not align smoothly with diabetes burden; the residual analysis identifies systematic over- and under-coverage relative to peer states, and sponsor composition indicates that much of the extra density in over-covered states is driven by industry rather than academic or federal research. At the county level, 73.4% of counties host no trial site and travel distances are often substantial. The modeling evidence (Aim 2) supports Hypothesis 2. Across three model families with matched tuning protocols, adding trial-access variables to an SES-only baseline produces at most a small improvement in cross-validated R^2. The feature-importance analyses — from both a linear and two non-linear perspectives — rank the trial-access variables below every major socioeconomic block.

Bigger-picture framing. Trial placement appears to be a consequence of institutional infrastructure and sponsor logistics rather than a response to local disease burden, and that placement is so correlated with the broader structural features of a county (poverty, insurance, education, racial composition) that, once those features are modeled, trial access adds little independent predictive information. This is a coherent — if uncomfortable — story: the same socioeconomic disadvantage that produces higher diabetes burden also excludes a county from the research infrastructure that develops new diabetes treatments. Improving trial access in high-burden, under-covered regions is therefore unlikely to be achieved by marginal adjustments to site selection alone; it requires investment in the upstream infrastructure (endocrinology workforce, academic medical centers, community health networks) that draws sponsors in the first place.

Limitations.

- The analyses are cross-sectional and ecological: inferences describe county-level associations, not individual-level causal relationships. A county with apparent access gaps may still have residents enrolled in trials elsewhere, and vice versa.
- Trial-access features are measured from a cumulative ClinicalTrials.gov snapshot, while burden and SES variables reflect 2018–2022 windows. Historical placement weighs into the cumulative count.
- County-level trial density is zero-inflated (73.4% of counties), so most of the access information lives in the log-distance feature. Haversine distance is also a crude proxy for real-world travel (road networks, drive time, and transit are not modeled).
- Medicaid expansion, rurality, and academic-center presence enter the analysis as state-level or binary county-level proxies; a finer county-level measure of research infrastructure (for example, proximity to a CTSA-hubbed academic medical center) might shift the feature-importance rankings.
- The model-comparison table in this report reflects one random 5-fold CV seed; variance across seeds was not reported here. The qualitative conclusion (small ΔR^2, SES dominates) is robust, but precise effect magnitudes should be interpreted with that caveat in mind.

Takeaway. The study set out to ask whether clinical-trial access varies independently of socioeconomic structure, and whether it carries any additional predictive signal for county-level diabetes burden beyond that structure. The answer — descriptively and predictively — is that trial access is largely a downstream reflection of the same structural factors that shape disease burden. Equity interventions on trial placement that ignore the underlying research-infrastructure gap are therefore likely to have small effects.

References

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